CASE STUDY #8 — TELECOM

SignalNet's Urban Network Congestion Crisis

Capacity Planning & Network Operations

Introduction

SignalNet, a metro-focused operator with 12 million subscribers, saw data speeds collapse during peak hours (7–10 PM) across its top 20 revenue-generating cities, even though the operator's headline network-coverage maps showed "full 4G/5G availability."

Background

Customers reported buffering on video calls and app timeouts. Churn in the affected circles rose from 1.8% to 4.1% monthly within one quarter, concentrated among high-ARPU postpaid users. Investigation revealed that cell-site capacity upgrades were triggered by average utilization rather than peak-hour utilization, the network had no dynamic load-balancing between overlapping 4G and 5G layers, KPI dashboards reported city-wide averages that masked hyper-local congestion, and network performance data was never linked to customer complaint or churn data.

Approach: OSS Concepts Applied

OSS Function	Issue Identified	Fix Implemented
Capacity Management	Upgrades triggered on average, not peak, utilization	Shifted trigger thresholds to 95th-percentile peak-hour utilization
Traffic Engineering / SON	No dynamic load balancing across 4G/5G layers	Deployed Self-Organizing Network (SON) load-balancing between carriers
Performance Management (FCAPS)	City-level KPI averaging hid local hotspots	Introduced grid-level (500m) congestion heatmaps
Customer Experience Management	No link between network KPIs and churn/complaints	Built a CEM layer correlating cell-level performance with complaint and churn data
Site Planning / Rollout	New site approvals were slow and reactive	Adopted a rolling 90-day proactive densification plan for top-20 revenue circles

Resolution & Outcome

Within one quarter of implementing the changes:

- Peak-hour data speeds in the top 20 circles improved by 38%
- Postpaid churn in affected circles fell from 4.1% back to 2.0%
- Congestion-related complaint volume dropped by 42%

Final Note

The crisis was rooted in a measurement blind spot, not a raw capacity shortfall: averaged KPIs hid the peak-hour, hyper-local congestion that customers actually experienced. Fixing what engineering measured — and linking it to churn — fixed the underlying problem.

Questions for Discussion

- Why can a network appear "fully covered" on paper while still failing customers at the exact moments that matter most?
- What are the trade-offs of triggering capacity upgrades on peak-hour utilization versus average utilization?

- How should SignalNet prioritize investment between the top 20 revenue circles and its broader network footprint going forward?